Product Requirement Document (PRD)

Project Name: Valence

Track: AI Data Governance (TrojanHacks 2026)

Team: Valence

Core Thesis: *The environment carries the signal, not the face.* Traditional affective computing is context-blind and relies on intrusive, non-consensual biometric harvesting. **Valence** is a privacy-preserving, multimodal contextual AI agent that reads the room and the behavior—not just a isolated face—to accurately diagnose and explain human emotional states.

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1. Executive Summary & Problem Statement

The Problem

Current emotion recognition AI faces critical hurdles that stall its real-world accuracy and ethics:

- **The Masking Dilemma:** Models map surface expressions, misinterpreting a forced, polite smile as genuine happiness while ignoring underlying frustration.
- **Context Blindness:** A person with their head in their hands could be weeping, laughing, or deeply focused. Without environmental, temporal, or behavioral context, deep learning models default to false conclusions.
- **The Black Box & Privacy Nightmare:** Extractive biometric pipelines continuously scrape and store Personally Identifiable Information (PII) without continuous consent. They output arbitrary emotion labels with zero transparency into *why* the model made that decision.

The Solution

Valence is a hybrid individual-and-environmental tracking system. It processes live video and audio streams at the edge to detect macro environmental stimuli (noise, light, time dynamics) alongside non-PII somatic mechanics (clenching fists, postural collapses) and social triggers (compliments, keyword indicators).

Crucially, **Valence** enforces structural data governance: it extracts safe, anonymous numerical event counts, immediately deletes raw video/audio frames from RAM, and compiles these metrics into a **Rolling State Vector**. The system uses this contextual timeline to accurately classify emotional states and outputs a natural language audit log explaining the exact context behind its predictions.

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2. System Architecture & Data Lifecycle

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[In-the-Wild Stream] ──> [Local Edge Inference] ──> [Biometric Purge Gate] ──> [Rolling History Window] ──> [Explainable Prediction Engine]
(Audio, Video, Lux)       (Count Behaviors/Lux)     (Wipe Raw Frame & Audio)      (15-Minute State Vector)      (Label + Explainable Audit Log)
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Step 1: Ingestion & Data Minimization (The Governance Gate)

The system ingests raw camera, microphone, and ambient light data. Instead of caching or transmitting this data, local edge modules extract anonymous event metadata and **instantly purge** the raw streams from memory:

- *Video stream* is processed locally to extract body mechanics and individual bounding box locations, then immediately deleted.
- *Audio stream* is processed for average amplitude and a keyword match (laughter, phrase matching), then immediately deleted.

Step 2: The Multimodal Rolling State Vector Schema

The extracted metadata is pushed to a thread-safe sliding queue representing a rolling **15-minute window** of the environment. The compiled vector tracks three distinct contextual layers:

Layer	Captured Metrics	Purpose
Ambient Physics	Ambient Noise (dB), Lighting Intensity (lux)	Tracks baseline physical stressors in the room.

Time Dynamics	Hours since last break, Time of day	Models exhaustion fatigue curves over time.
Somatic & Social Mechanics	Clenched fist counts, posture slumps, keyword triggers (compliments, laughter)	Tracks micro-behaviors that reveal inner states or social interactions.

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3. Product Features & User Interface (UI) Scope

The user interface acts as an **Explainable AI (XAI) Audit Trail** for judges and users to verify compliance and prediction accuracy.

Live Video Monitor & Overlay

- Displays the live webcam feed with temporary bounding box overlays over individuals currently in frame (e.g., `Person #1`).
- **Governance Visual Anchor:** A prominent terminal log or flashing indicator displaying: `[GOVERNANCE] Raw biometric frame purged successfully. Appended state vector to timeline.`

The Explainable AI (XAI) Description Box

Directly beneath the live monitor, the interface prints the predicted emotion label accompanied by a transparent breakdown of the data governance logic, solving the "Black Box" problem:

- **Detected State:** `Person #1: Frustrated / Masked Stress`
- **Context-Aware Audit Explanation:**
  "Model predicted 'Frustrated' with 88% confidence. While Person #1 is currently displaying a superficial smiling expression (high facial valence), the system detected a somatic trigger (clenched fists) paired with a negative environmental history layer: high ambient noise (72dB) and an extended duration since the last communal room break (4.2 hours). Context corrected the facial mask."

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4. Sunday Hackathon Test Plan & Pitch Strategy

To win the **AI Data Governance** vertical, the team will present three core proofs on Sunday morning:

1. **The Proof of Data Minimization (The Code):** Presenting the pipeline code snippet to the judges, highlighting the explicit `del frame` and memory purge functionality that prevents PII storage.
2. **The Context vs. Face Experiment (The Evaluation):** Presenting a comparative chart proving our core hypothesis: that the rolling environmental/behavioral vector yields **higher predictive accuracy over time** in-the-wild than a context-blind facial snapshot algorithm.
3. **The Anti-Bias Validation:** Demonstrating how our system prevents demographic or lighting bias—even if a face is partially obscured or poorly lit, the tracking of ambient physics, time dynamics, and alternative somatic triggers ensures stable performance.